



The University of the State of New York
REGENTS HIGH SCHOOL EXAMINATION

ALGEBRA II

(Common Core)

Friday, June 16, 2017 — 1:15 to 4:15 p.m., only

Student Name _____

School Name _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for **Part I** has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 37 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in **Parts II, III, and IV** directly in this booklet. All work should be written in pen, except for graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will not be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice ...

A graphing calculator and a straightedge (ruler) must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

High School Math Reference Sheet

1 inch = 2.54 centimeters	1 kilometer = 0.62 mile	1 cup = 8 fluid ounces
1 meter = 39.37 inches	1 pound = 16 ounces	1 pint = 2 cups
1 mile = 5280 feet	1 pound = 0.454 kilogram	1 quart = 2 pints
1 mile = 1760 yards	1 kilogram = 2.2 pounds	1 gallon = 4 quarts
1 mile = 1.609 kilometers	1 ton = 2000 pounds	1 gallon = 3.785 liters
		1 liter = 0.264 gallon
		1 liter = 1000 cubic centimeters

Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Circle	$A = \pi r^2$
Circle	$C = \pi d$ or $C = 2\pi r$
General Prisms	$V = Bh$
Cylinder	$V = \pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$
Cone	$V = \frac{1}{3}\pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$

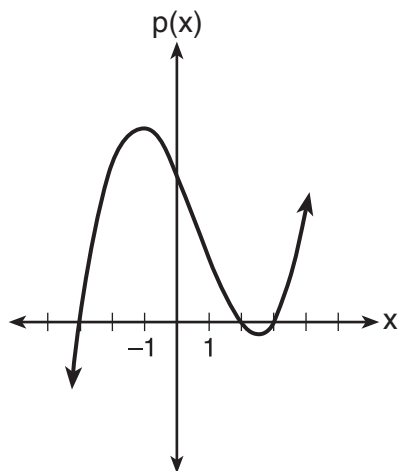
Pythagorean Theorem	$a^2 + b^2 = c^2$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Arithmetic Sequence	$a_n = a_1 + (n - 1)d$
Geometric Sequence	$a_n = a_1 r^{n-1}$
Geometric Series	$S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$
Radians	1 radian = $\frac{180}{\pi}$ degrees
Degrees	1 degree = $\frac{\pi}{180}$ radians
Exponential Growth/Decay	$A = A_0 e^{k(t - t_0)} + B_0$

Part I

Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

Use this space for
computations.

- 1 The graph of the function $p(x)$ is sketched below.



Which equation could represent $p(x)$?

- (1) $p(x) = (x^2 - 9)(x - 2)$
 - (2) $p(x) = x^3 - 2x^2 + 9x + 18$
 - (3) $p(x) = (x^2 + 9)(x - 2)$
 - (4) $p(x) = x^3 + 2x^2 - 9x - 18$
- 2 What is the solution to $8(2^{x+3}) = 48$?
- (1) $x = \frac{\ln 6}{\ln 2} - 3$
 - (2) $x = 0$
 - (3) $x = \frac{\ln 48}{\ln 16} - 3$
 - (4) $x = \ln 4 - 3$

3 Cheap and Fast gas station is conducting a consumer satisfaction survey. Which method of collecting data would most likely lead to a biased sample?

- (1) interviewing every 5th customer to come into the station
- (2) interviewing customers chosen at random by a computer at the checkout
- (3) interviewing customers who call an 800 number posted on the customers' receipts
- (4) interviewing every customer who comes into the station on a day of the week chosen at random out of a hat

4 The expression $6xi^3(-4xi + 5)$ is equivalent to

- (1) $2x - 5i$
- (2) $-24x^2 - 30xi$
- (3) $-24x^2 + 30x - i$
- (4) $26x - 24x^2i - 5i$

5 If $f(x) = 3|x| - 1$ and $g(x) = 0.03x^3 - x + 1$, an approximate solution for the equation $f(x) = g(x)$ is

- (1) 1.96
- (2) 11.29
- (3) $(-0.99, 1.96)$
- (4) $(11.29, 32.87)$

6 Given the parent function $p(x) = \cos x$, which phrase best describes the transformation used to obtain the graph of $g(x) = \cos(x + a) - b$, if a and b are positive constants?

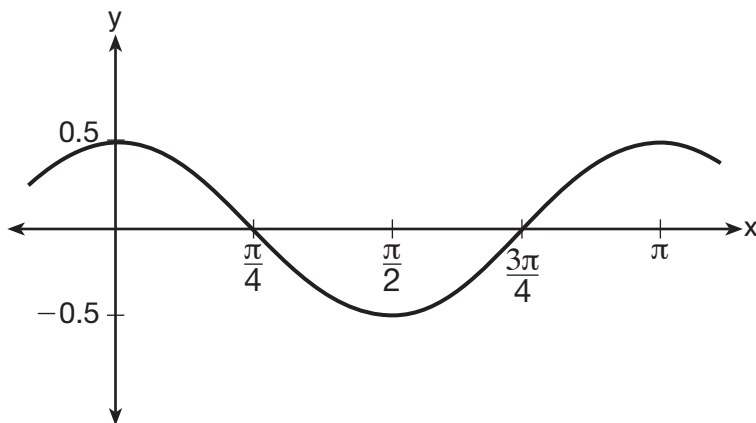
- (1) right a units, up b units
- (2) right a units, down b units
- (3) left a units, up b units
- (4) left a units, down b units

**Use this space for
computations.**

7 The solution to the equation $4x^2 + 98 = 0$ is

- (1) ± 7 (3) $\pm \frac{7\sqrt{2}}{2}$
(2) $\pm 7i$ (4) $\pm \frac{7i\sqrt{2}}{2}$

8 Which equation is represented by the graph shown below?



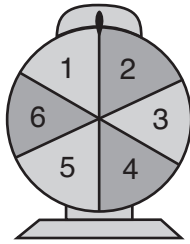
- (1) $y = \frac{1}{2} \cos 2x$ (3) $y = \frac{1}{2} \cos x$
(2) $y = \cos x$ (4) $y = 2 \cos \frac{1}{2}x$

9 A manufacturing company has developed a cost model, $C(x) = 0.15x^3 + 0.01x^2 + 2x + 120$, where x is the number of items sold, in thousands. The sales price can be modeled by $S(x) = 30 - 0.01x$. Therefore, revenue is modeled by $R(x) = x \cdot S(x)$.

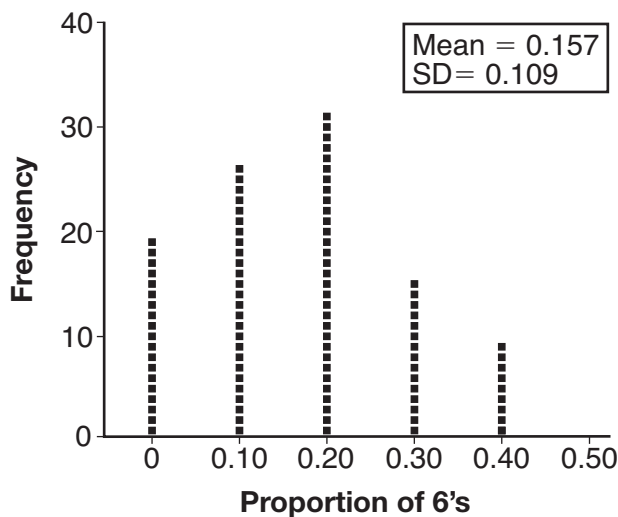
The company's profit, $P(x) = R(x) - C(x)$, could be modeled by

- (1) $0.15x^3 + 0.02x^2 - 28x + 120$
(2) $-0.15x^3 - 0.02x^2 + 28x - 120$
(3) $-0.15x^3 + 0.01x^2 - 2.01x - 120$
(4) $-0.15x^3 + 32x + 120$

- 10 A game spinner is divided into 6 equally sized regions, as shown in the diagram below.



For Miles to win, the spinner must land on the number 6. After spinning the spinner 10 times, and losing all 10 times, Miles complained that the spinner is unfair. At home, his dad ran 100 simulations of spinning the spinner 10 times, assuming the probability of winning each spin is $\frac{1}{6}$. The output of the simulation is shown in the diagram below.



Which explanation is appropriate for Miles and his dad to make?

- (1) The spinner was likely unfair, since the number 6 failed to occur in about 20% of the simulations.
- (2) The spinner was likely unfair, since the spinner should have landed on the number 6 by the sixth spin.
- (3) The spinner was likely not unfair, since the number 6 failed to occur in about 20% of the simulations.
- (4) The spinner was likely not unfair, since in the output the player wins once or twice in the majority of the simulations.

11 Which binomial is a factor of $x^4 - 4x^2 - 4x + 8$?

- (1) $x - 2$ (3) $x - 4$
(2) $x + 2$ (4) $x + 4$

12 Given that $\sin^2 \theta + \cos^2 \theta = 1$ and $\sin \theta = -\frac{\sqrt{2}}{5}$, what is a possible value of $\cos \theta$?

- (1) $\frac{5 + \sqrt{2}}{5}$ (3) $\frac{3\sqrt{3}}{5}$
(2) $\frac{\sqrt{23}}{5}$ (4) $\frac{\sqrt{35}}{5}$

13 A student studying public policy created a model for the population of Detroit, where the population decreased 25% over a decade. He used the model $P = 714(0.75)^d$, where P is the population, in thousands, d decades after 2010. Another student, Suzanne, wants to use a model that would predict the population after y years. Suzanne's model is best represented by

- (1) $P = 714(0.6500)^y$ (3) $P = 714(0.9716)^y$
(2) $P = 714(0.8500)^y$ (4) $P = 714(0.9750)^y$

14 The probability that Gary and Jane have a child with blue eyes is 0.25, and the probability that they have a child with blond hair is 0.5. The probability that they have a child with both blue eyes and blond hair is 0.125. Given this information, the events blue eyes and blond hair are

- I: dependent
II: independent
III: mutually exclusive

- (1) I, only (3) I and III
(2) II, only (4) II and III

- 15** Based on climate data that have been collected in Bar Harbor, Maine, the average monthly temperature, in degrees F, can be modeled by the equation $B(x) = 23.914\sin(0.508x - 2.116) + 55.300$. The same governmental agency collected average monthly temperature data for Phoenix, Arizona, and found the temperatures could be modeled by the equation $P(x) = 20.238\sin(0.525x - 2.148) + 86.729$.

Which statement can *not* be concluded based on the average monthly temperature models x months after starting data collection?

- (1) The average monthly temperature variation is more in Bar Harbor than in Phoenix.
- (2) The midline average monthly temperature for Bar Harbor is lower than the midline temperature for Phoenix.
- (3) The maximum average monthly temperature for Bar Harbor is 79° F, to the nearest degree.
- (4) The minimum average monthly temperature for Phoenix is 20° F, to the nearest degree.

- 16** For $x \neq 0$, which expressions are equivalent to one divided by the sixth root of x ?

$$\text{I. } \frac{\sqrt[6]{x}}{\sqrt[3]{x}} \quad \text{II. } \frac{x^{\frac{1}{6}}}{x^{\frac{1}{3}}} \quad \text{III. } x^{-\frac{1}{6}}$$

- (1) I and II, only
- (2) I and III, only
- (3) II and III, only
- (4) I, II, and III

- 17** A parabola has its focus at (1,2) and its directrix is $y = -2$. The equation of this parabola could be

- (1) $y = 8(x + 1)^2$
- (2) $y = \frac{1}{8}(x + 1)^2$
- (3) $y = 8(x - 1)^2$
- (4) $y = \frac{1}{8}(x - 1)^2$

- 18** The function $p(t) = 110e^{0.03922t}$ models the population of a city, in millions, t years after 2010. As of today, consider the following two statements:

- I. The current population is 110 million.
- II. The population increases continuously by approximately 3.9% per year.

This model supports

- (1) I, only
- (2) II, only
- (3) both I and II
- (4) neither I nor II

- 19** To solve $\frac{2x}{x-2} - \frac{11}{x} = \frac{8}{x^2-2x}$, Ren multiplied both sides by the least common denominator. Which statement is true?

- (1) 2 is an extraneous solution.
- (2) $\frac{7}{2}$ is an extraneous solution.
- (3) 0 and 2 are extraneous solutions.
- (4) This equation does not contain any extraneous solutions.

- 20** Given $f(9) = -2$, which function can be used to generate the sequence $-8, -7.25, -6.5, -5.75, \dots$?

- (1) $f(n) = -8 + 0.75n$
- (2) $f(n) = -8 - 0.75(n - 1)$
- (3) $f(n) = -8.75 + 0.75n$
- (4) $f(n) = -0.75 + 8(n - 1)$

- 21** The function $f(x) = 2^{-0.25x} \cdot \sin\left(\frac{\pi}{2}x\right)$ represents a damped sound wave function. What is the average rate of change for this function on the interval $[-7, 7]$, to the nearest hundredth?

- (1) -3.66
- (2) -0.30
- (3) -0.26
- (4) 3.36

- 22** Mallory wants to buy a new window air conditioning unit. The cost for the unit is \$329.99. If she plans to run the unit three months out of the year for an annual operating cost of \$108.78, which function models the cost per year over the lifetime of the unit, $C(n)$, in terms of the number of years, n , that she owns the air conditioner?

(1) $C(n) = 329.99 + 108.78n$

(2) $C(n) = 329.99 + 326.34n$

(3) $C(n) = \frac{329.99 + 108.78n}{n}$

(4) $C(n) = \frac{329.99 + 326.34n}{n}$

- 23** The expression $\frac{-3x^2 - 5x + 2}{x^3 + 2x^2}$ can be rewritten as

(1) $\frac{-3x - 3}{x^2 + 2x}$

(3) $-3x^{-1} + 1$

(2) $\frac{-3x - 1}{x^2}$

(4) $-3x^{-1} + x^{-2}$

- 24** Jasmine decides to put \$100 in a savings account each month. The account pays 3% annual interest, compounded monthly. How much money, S , will Jasmine have after one year?

(1) $S = 100(1.03)^{12}$

(3) $S = 100(1.0025)^{12}$

(2) $S = \frac{100 - 100(1.0025)^{12}}{1 - 1.0025}$

(4) $S = \frac{100 - 100(1.03)^{12}}{1 - 1.03}$

Part II

Answer all 8 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

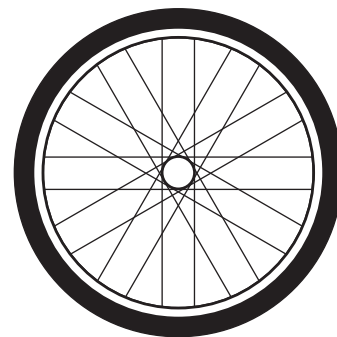
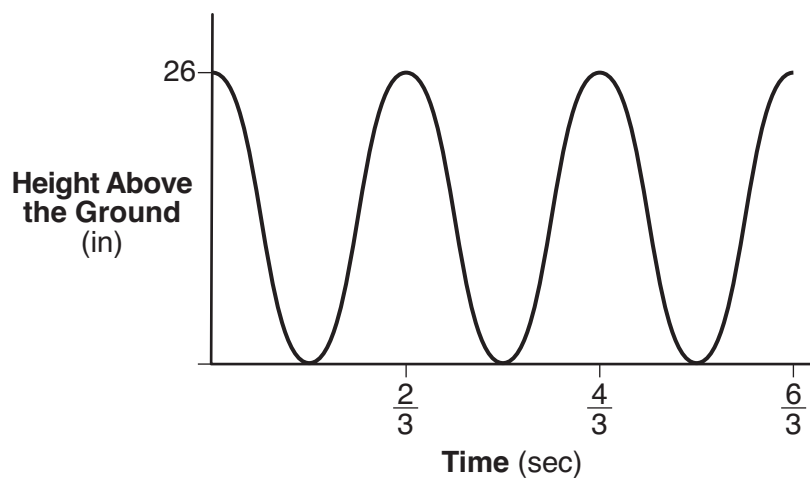
25 Given $r(x) = x^3 - 4x^2 + 4x - 6$, find the value of $r(2)$.

What does your answer tell you about $x - 2$ as a factor of $r(x)$? Explain.

26 The weight of a bag of pears at the local market averages 8 pounds with a standard deviation of 0.5 pound. The weights of all the bags of pears at the market closely follow a normal distribution. Determine what percentage of bags, to the *nearest integer*, weighed *less* than 8.25 pounds.

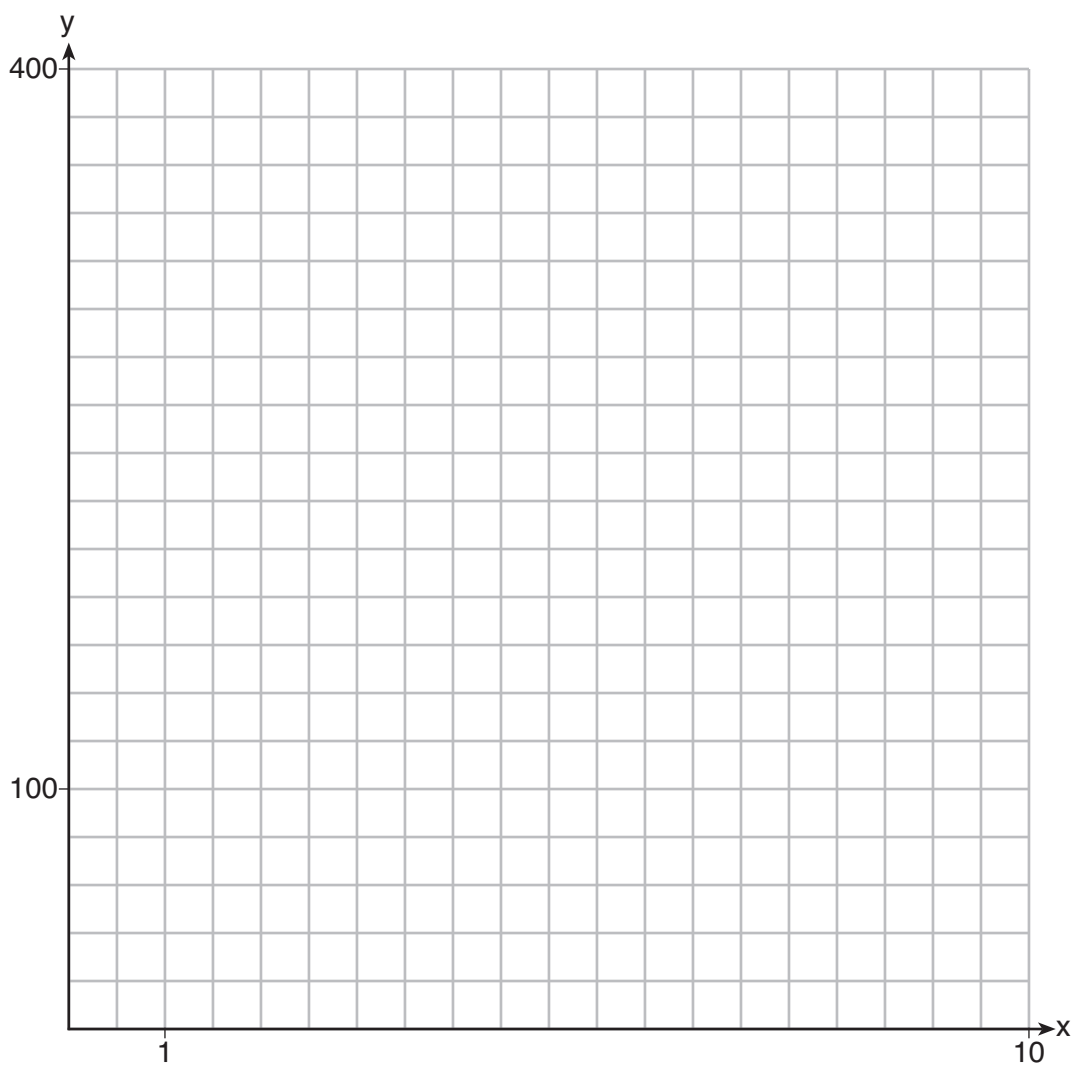
27 Over the set of integers, factor the expression $4x^3 - x^2 + 16x - 4$ completely.

- 28** The graph below represents the height above the ground, h , in inches, of a point on a triathlete's bike wheel during a training ride in terms of time, t , in seconds.



Identify the period of the graph and describe what the period represents in this context.

29 Graph $y = 400(.85)^{2x} - 6$ on the set of axes below.



30 Solve algebraically for all values of x :

$$\sqrt{x-4} + x = 6$$

31 Write $\sqrt[3]{x} \cdot \sqrt{x}$ as a single term with a rational exponent.

32 Data collected about jogging from students with two older siblings are shown in the table below.

	Neither Sibling Jogs	One Sibling Jogs	Both Siblings Jog
Student Does Not Jog	1168	1823	1380
Student Jogs	188	416	400

Using these data, determine whether a student with two older siblings is more likely to jog if one sibling jogs or if both siblings jog. Justify your answer.

Part III

Answer all 4 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

33 Solve the following system of equations algebraically for all values of x , y , and z :

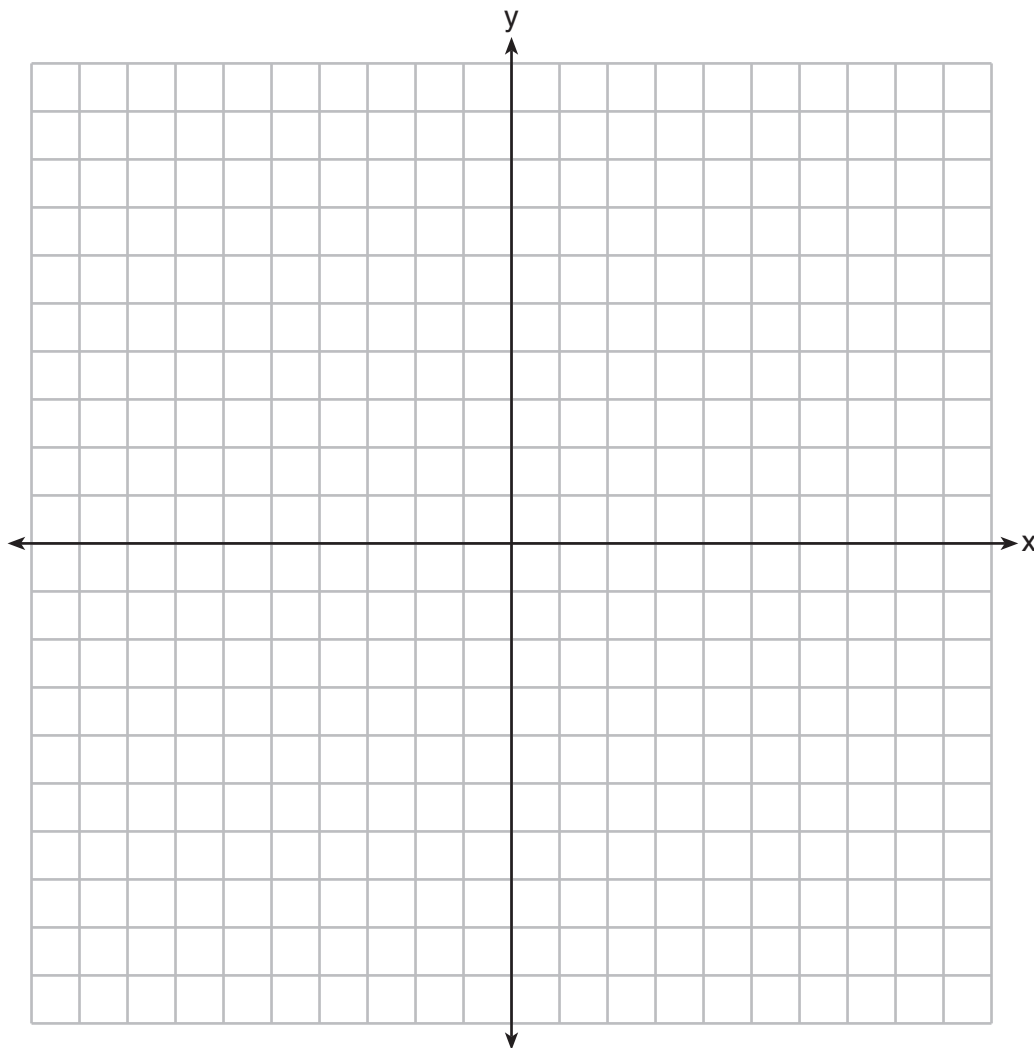
$$\begin{aligned}x + y + z &= 1 \\2x + 4y + 6z &= 2 \\-x + 3y - 5z &= 11\end{aligned}$$

34 Jim is looking to buy a vacation home for \$172,600 near his favorite southern beach. The formula to compute a mortgage payment, M , is $M = P \cdot \frac{r(1+r)^N}{(1+r)^N - 1}$ where P is the principal amount of the loan, r is the monthly interest rate, and N is the number of monthly payments. Jim's bank offers a monthly interest rate of 0.305% for a 15-year mortgage.

With no down payment, determine Jim's mortgage payment, rounded to the *nearest dollar*.

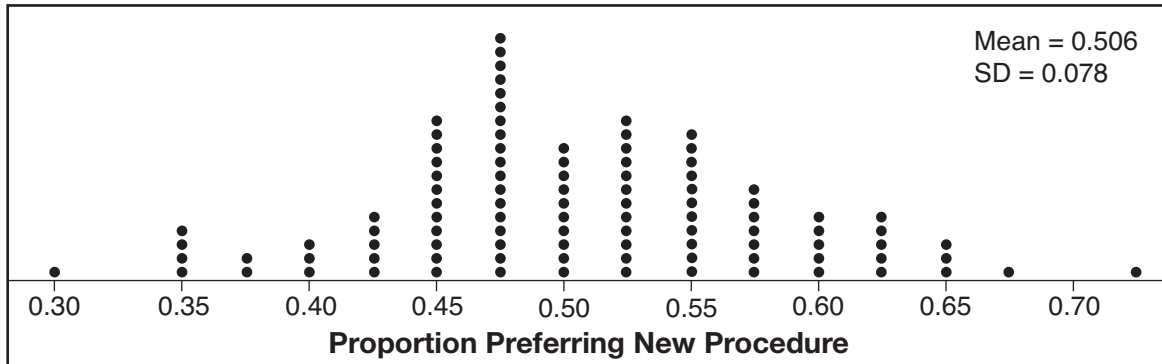
Algebraically determine and state the down payment, rounded to the *nearest dollar*, that Jim needs to make in order for his mortgage payment to be \$1100.

- 35** Graph $y = \log_2(x + 3) - 5$ on the set of axes below. Use an appropriate scale to include *both* intercepts.



Describe the behavior of the given function as x approaches -3 and as x approaches positive infinity.

- 36** Charlie's Automotive Dealership is considering implementing a new check-in procedure for customers who are bringing their vehicles for routine maintenance. The dealership will launch the procedure if 50% or more of the customers give the new procedure a favorable rating when compared to the current procedure. The dealership devises a simulation based on the minimal requirement that 50% of the customers prefer the new procedure. Each dot on the graph below represents the proportion of the customers who preferred the new check-in procedure, each of sample size 40, simulated 100 times.



Assume the set of data is approximately normal and the dealership wants to be 95% confident of its results. Determine an interval containing the plausible sample values for which the dealership will launch the new procedure. Round your answer to the *nearest hundredth*.

Forty customers are selected randomly to undergo the new check-in procedure and the proportion of customers who prefer the new procedure is 32.5%. The dealership decides *not* to implement the new check-in procedure based on the results of the study. Use statistical evidence to explain this decision.

Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided to determine your answer. Note that diagrams are not necessarily drawn to scale. A correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [6]

- 37 A radioactive substance has a mass of 140 g at 3 p.m. and 100 g at 8 p.m. Write an equation in the form $A = A_0 \left(\frac{1}{2}\right)^{\frac{t}{h}}$ that models this situation, where h is the constant representing the number of hours in the half-life, A_0 is the initial mass, and A is the mass t hours after 3 p.m.

Using this equation, solve for h , to the *nearest ten thousandth*.

Determine when the mass of the radioactive substance will be 40 g. Round your answer to the *nearest tenth of an hour*.

If the student’s responses for the multiple-choice questions are being hand scored prior to being scanned, the scorer must be careful not to make any marks on the answer sheet except to record the scores in the designated score boxes. Marks elsewhere on the answer sheet will interfere with the accuracy of the scanning.

Part I

Allow a total of 48 credits, 2 credits for each of the following.

(1).....1.....	(9).....2.....	(17).....4.....
(2).....1.....	(10).....3.....	(18).....2.....
(3).....3.....	(11).....1.....	(19).....1.....
(4).....2.....	(12).....2.....	(20).....3.....
(5).....2.....	(13).....3.....	(21).....3.....
(6).....4.....	(14).....2.....	(22).....3.....
(7).....4.....	(15).....4.....	(23).....4.....
(8).....1.....	(16).....4.....	(24).....2.....

Updated information regarding the rating of this examination may be posted on the New York State Education Department’s web site during the rating period. Check this web site at: <http://www.p12.nysed.gov/assessment/> and select the link “Scoring Information” for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

The Department is providing supplemental scoring guidance, the “Model Response Set,” for the Regents Examination in Algebra II (Common Core). This guidance is recommended to be part of the scorer training. Schools are encouraged to incorporate the Model Response Sets into the scorer training or to use them as additional information during scoring. While not reflective of all scenarios, the model responses selected for the Model Response Set illustrate how less common student responses to constructed-response questions may be scored. The Model Response Set will be available on the Department’s web site at <http://www.nysedregents.org/algebratwo/>.

**Map to the Common Core Learning Standards
Algebra II (Common Core)
June 2017**

Question	Type	Credits	Cluster
1	Multiple Choice	2	A-APR.B
2	Multiple Choice	2	F-LE.A
3	Multiple Choice	2	S-IC.B
4	Multiple Choice	2	N-CN.A
5	Multiple Choice	2	A-REI.D
6	Multiple Choice	2	F-BF.B
7	Multiple Choice	2	A-REI.B
8	Multiple Choice	2	F-IF.C
9	Multiple Choice	2	F-BF.A
10	Multiple Choice	2	S-IC.A
11	Multiple Choice	2	A-APR.B
12	Multiple Choice	2	F-TF.C
13	Multiple Choice	2	A-SSE.B
14	Multiple Choice	2	S-CP.B
15	Multiple Choice	2	F-IF.B
16	Multiple Choice	2	N-RN.A
17	Multiple Choice	2	G-GPE.A
18	Multiple Choice	2	F-LE.B
19	Multiple Choice	2	A-REI.A
20	Multiple Choice	2	F-IF.A

21	Multiple Choice	2	F-IF.B
22	Multiple Choice	2	A-CED.A
23	Multiple Choice	2	A-SSE.A
24	Multiple Choice	2	A-SSE.B
25	Constructed Response	2	A-APR.B
26	Constructed Response	2	S-ID.A
27	Constructed Response	2	A-SSE.A
28	Constructed Response	2	F-IF.B
29	Constructed Response	2	F-IF.C
30	Constructed Response	2	A-REI.A
31	Constructed Response	2	N-RN.A
32	Constructed Response	2	S-CP.A
33	Constructed Response	4	A-REI.C
34	Constructed Response	4	A-SSE.B
35	Constructed Response	4	F-IF.C
36	Constructed Response	4	S-IC.B
37	Constructed Response	6	F-BF.A

Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(25) [2] -6 , and a correct explanation is written.

[1] One computational error is made.

or

[1] One conceptual error is made.

or

[1] -6 , but the explanation is incorrect or missing.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(26) [2] 69 , and correct work is shown.

[1] Appropriate work is shown, but one computational error or rounding error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] 69 , but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(27) [2] $(x^2 + 4)(4x - 1)$ and correct work is shown.

[1] Appropriate work is shown, but one factoring error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] $x^2(4x - 1) + 4(4x - 1)$ is written, but no further correct work is shown.

or

[1] $(x^2 + 4)(4x - 1)$, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(28) [2] $\frac{2}{3}$ and a correct description is written.

[1] One computational error is made.

or

[1] One conceptual error is made.

or

[1] $\frac{2}{3}$, but the description is incomplete or missing.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(29) [2] A correct graph is drawn.

[1] One computational or graphing error is made.

or

[1] One conceptual error is made.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (30) [2] 5 and correct algebraic work is shown.
- [1] Appropriate work is shown, but one computational or simplification error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] Appropriate work is shown, but 8 is not rejected.
- or**
- [1] 5, but a method other than algebraic is used.
- or**
- [1] 5, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
-
- (31) [2] $x^{\frac{5}{6}}$ and correct work is shown.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] $x^{\frac{5}{6}}$, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
-
- (32) [2] Both siblings jog and a correct justification is given.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- [0] Both siblings jog, but no justification is given.
- or**
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(33) [4] $x = 0$, $y = 2$, and $z = -1$, and correct algebraic work is shown.

[3] Appropriate work is shown, but one computational error is made.

or

[3] Appropriate work is shown to find two solutions, but no further correct work is shown.

[2] Appropriate work is shown, but two or more computational errors are made.

or

[2] Appropriate work is shown, but one conceptual error is made.

or

[2] Appropriate work is shown to find one solution, but no further correct work is shown.

or

[2] $x = 0$, $y = 2$, and $z = -1$, but a method other than algebraic is used.

[1] Appropriate work is shown, but one conceptual error and one computational error are made.

or

[1] Appropriate work is shown to eliminate a variable in order to create a system of two equations, but no further correct work is shown.

or

[1] $x = 0$, $y = 2$, and $z = -1$, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (34) [4] 1247 and 20,407, and correct work is shown.
- [3] Appropriate work is shown, but one computational or rounding error is made.
- or**
- [3] Appropriate work is shown to find 1247 and 152,193, but no further correct work is shown.
- [2] Appropriate work is shown, but two or more computational or rounding errors are made.
- or**
- [2] Appropriate work is shown, but one conceptual error is made.
- or**
- [2] Appropriate work is shown to find 1247 or 20,407, but no further correct work is shown.
- or**
- [2] 1247 and 20,407, but no work is shown.
- [1] Appropriate work is shown, but one conceptual error and one computational or rounding error are made.
- or**
- [1] A correct substitution is made in the mortgage payment formula, but no further correct work is shown.
- or**
- [1] 1247 or 20,407, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (35) [4] A correct graph is drawn and a correct description is given, such as $x \rightarrow -3$, $y \rightarrow -\infty$, and as $x \rightarrow \infty$, $y \rightarrow \infty$.
- [3] Appropriate work is shown, but one computational or graphing error is made.
- [2] Appropriate work is shown, but two or more computational or graphing errors are made.
- or**
- [2] Appropriate work is shown, but one conceptual error is made.
- or**
- [2] A correct description is given, but no further correct work is shown.
- or**
- [2] A correct graph is drawn, but no further correct work is shown.
- [1] Appropriate work is shown, but one conceptual error and one computational or graphing error are made.
- or**
- [1] As $x \rightarrow -3$, $y \rightarrow -\infty$, or as $x \rightarrow \infty$, $y \rightarrow \infty$.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (36) [4] A correct interval is stated, such as $(0.35, 0.66)$, and correct work is shown, and a correct explanation is written.
- [3] Appropriate work is shown, but one computational or rounding error is made.
- or**
- [3] Appropriate work is shown, but the sample proportion is used to find the interval.
- [2] Appropriate work is shown, but two or more computational or rounding errors are made.
- or**
- [2] Appropriate work is shown, but one conceptual error is made.
- or**
- [2] A correct interval, and correct work is shown, but no further correct work is shown.
- or**
- [2] A correct explanation is written, but no further correct work is shown.
- [1] Appropriate work is shown but one conceptual error and one computational or rounding error are made.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
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Part IV

For each question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

- (37) [6] $100 = 140\left(\frac{1}{2}\right)^{\frac{5}{h}}$ or equivalent, 10.3002, 18.6, and correct work is shown.
- [5] Appropriate work is shown, but one computational, rounding, or notation error is made.
- [4] Appropriate work is shown, but two computational or rounding errors are made.
- or**
- [4] Appropriate work is shown, but one conceptual error is made.
- [3] Appropriate work is shown, but one conceptual error and one computational error are made.
- or**
- [3] Appropriate work is shown to find 10.3002, but no further correct work is shown.
- [2] Appropriate work is shown, but one conceptual error and two or more computational or rounding errors are made.
- or**
- [2] Appropriate work is shown, but two conceptual errors are made.
- or**
- [2] Appropriate work is shown to find 18.6, but no further correct work is shown.
- or**
- [2] 10.3002 and 18.6, but no work is shown.
- [1] $100 = 140\left(\frac{1}{2}\right)^{\frac{5}{h}}$, but no further correct work is shown.
- or**
- [1] 10.3002 or 18.6, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
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